Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMIDS36523
Project Name	Empowering India: analysing the evaluation of union budget allocation
Maximum Marks	Marks

Technical Architecture:

The technology stack for the Union Budget Analysis project includes collecting government budget datasets stored in a relational database like MySQL or PostgreSQL. Python (Pandas, NumPy) is used for data cleaning, transformation, and analysis, while calculated fields help generate insights such as totals, growth percentages, and top-performing schemesThe final interactive dashboard is built using Power BI or Tableau and deployed on a cloud platform for accessibility, performance, and scalability.

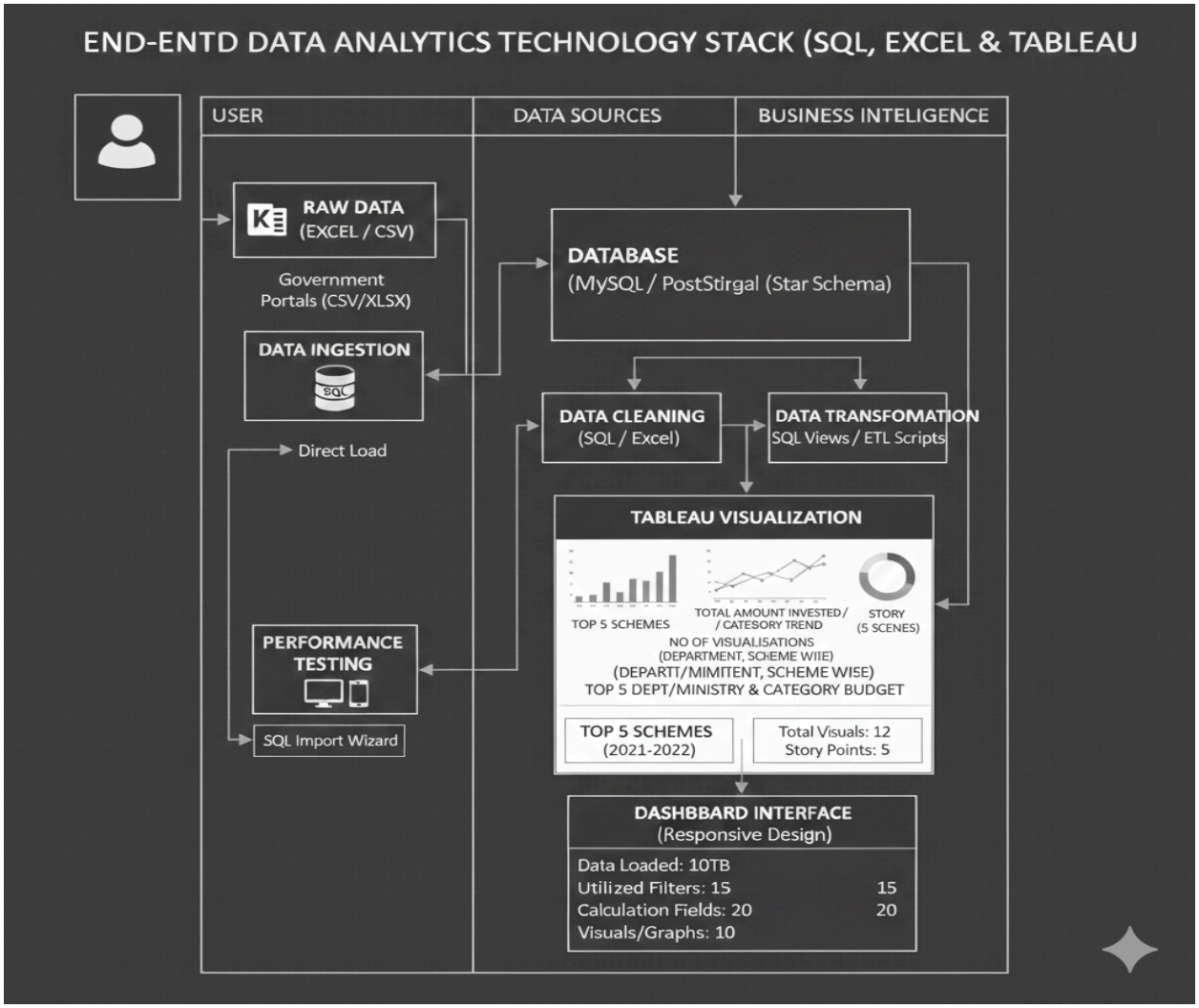

Table-1 : Components & Technologies:

SI. No	Component	Description	Technology
1	Data Source	Collect Union Budget data (2021–2024)	Kaggele website (CSV/EXCEL)
2	Data Ingestion	Import raw datasets into system	Excel/text file
3	Database	Store structured data	Tableau/Excel
4	Data Cleaning	Remove inconsistencies & errors	Tableau/Excel(filters)
5	Data Transformation	Structure data for analysis	Tableau/Excel

6	Data Modeling	Define relationships & schema	Tableau/Excel (filters)
7	Analytics Engine	Perform calculations & KPIs	Tableau
8	Visualization Tool	Build interactive dashboard	Tableau
9	Analytics Layer	KPI calculations (Top 5, Growth %, Totals)	Tableau
10	Deployment	Publish & share dashboard and Story	Tableau

Table-2: Application Characteristics:

SI. No	Characteristics	Description	Technology
1	Usability	Interactive dashboard with filters and slicers for easy navigation	Tableau
2	Performance	Fast data processing and optimized queries	SQL
3	Scalability	Supports additional years and increased data volume	SQl/Excel

4	Reliability	Cleaned and validated structured dataset	Tableau/ Excel
5	Security	Role-based access and secure data handling	Tableau (database security)
6	Availability	24/7 online dashboard access	Tableau public
7	Maintainability	Easy updates and modular architecture	Python + Database Schema
8	Analytical capability	Growth %, Top 5 analysis, category comparison	Calculated Fields
9	Responsiveness	Mobile-friendly dashboard design	Tableau